Category

The Term Structure of Interest Rates: Spot, Par, and Forward Curves

SPOT, PAR, AND FORWARD YIELD CURVES AND INTERPRETING THEIR RELATIONSHIP

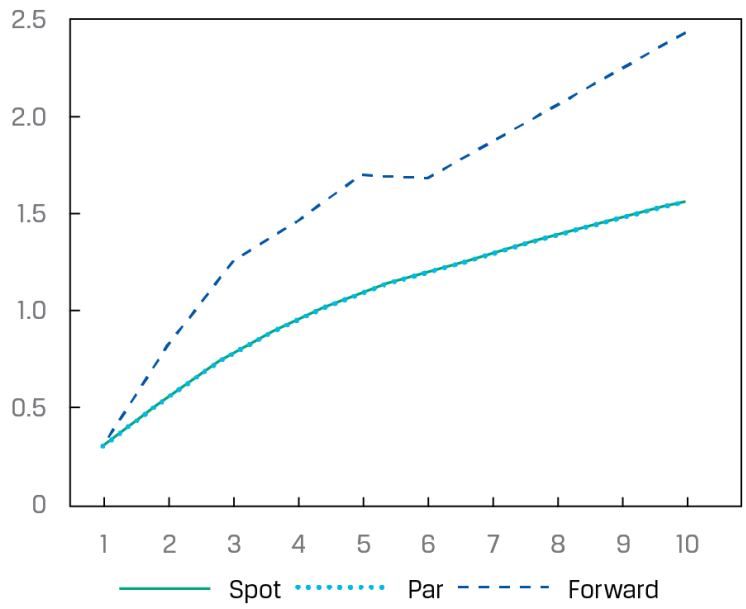
Learning Outcome

- compare the spot curve, par curve, and forward curve

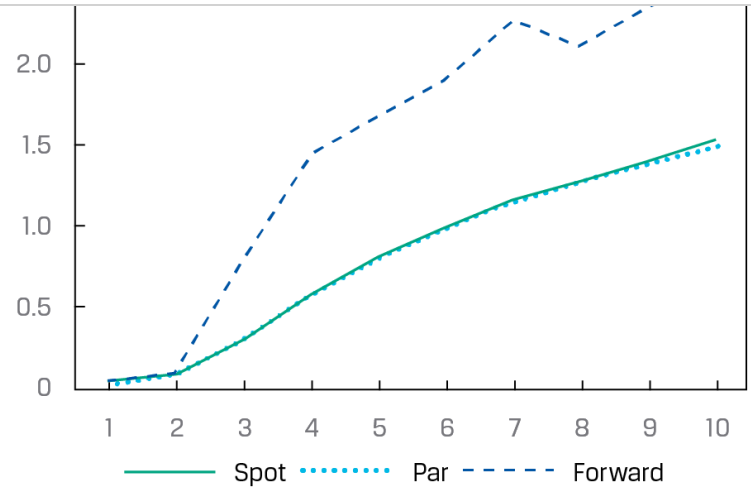
Because spot, par, and forward rates are interconnected, so are the shapes of the spot, par, and forward curves. This analysis is best illustrated through an example. In an earlier lesson, we used spot rates to compute par rates and forward rates for Canadian and Australian government bonds. We can use these rates to build yield curves across a range of maturities, shown in Exhibit 1.

Spot, Par, and Forward Curves (Canada and Australia)

A. Spot, Par, and Forward Curves: Canadian Government Bonds



B. Spot, Par, and Forward Curves: Australian Government Bonds



We can make the following general observations:

1. The spot rates are positive, and the spot curve is upward sloping.
2. The spot and par curves are nearly identical; the par rates are slightly lower than the spot rates, and the (slight) difference between the spot and par curves is greater at longer maturities.
3. Forward rates are greater than the spot and par rates.

These observations stem from general relationships between the shapes of the curves. If the spot curve is upward sloping, par rates will be near—but below—spot rates, particularly at the long end of the curve. This is because the low short-term spot rates result in higher bond prices, particularly for longer-term bonds, which results in low par rates when we do the calculation that assumes the price is 100% of par. Forward rates being above spot rates, if the spot curve is upward sloping, follows from the prior lesson, which established forward rates as incremental rates of return.

Exhibit 6 shows the spot, par, and forward curves if the spot curve is flat, assumed to be 2.50% for all maturities. When the spot rate is constant across all maturities, the par and forward rates will equate to the spot rate across all maturities as well. In short, a flat spot rate curve reflects no expectations of changes in future interest rates; thus forward rates equal spot rates. In this particular example, government bonds would pay a 2.5% coupon rate, and the bonds would be priced at par value (thus, the par rate is also 2.5%).

Exhibit 6:

Flat Term Structure

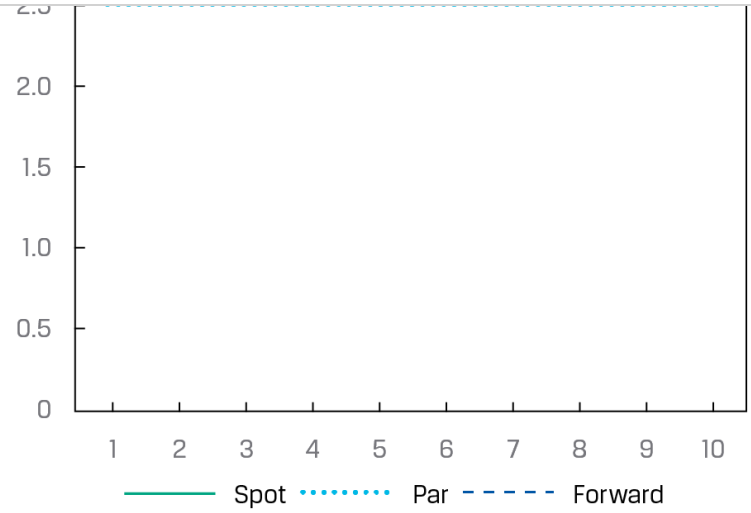
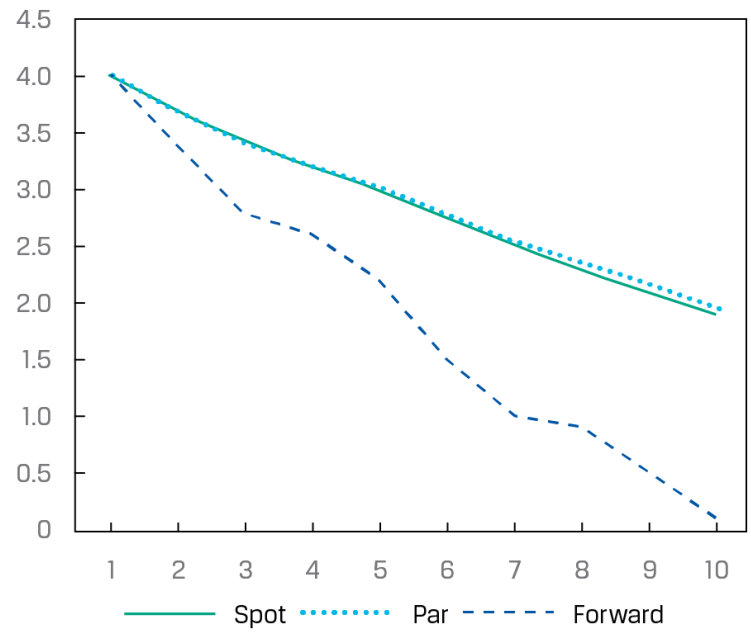


Exhibit 7 shows the spot, par, and forward curves for a downward-sloping (inverted) spot curve or falling spot rates as maturities lengthen. Starting from a 4% 1-year spot rate, rates fall to 1.90% for a maturity of 10 years. As in the previous examples, the par rates are similar to the spot rates. The inverted term structure of spot rates reflects expectations of significantly lower one-year rates in future years, and this is shown by the forward curve. In this example, the spot rate term structure implies a one-year rate of 0.1175% expected in nine years.

Exhibit 7:

Inverted Term Structure



We conclude with Exhibit 8, which summarizes the general relationship between the spot, par, and forward curves for different spot curve shapes.

Exhibit 8:

Spot, Par, and Forward Curve Relationship

Downward Sloping (Inverted) | Above spot curve | Below spot curve

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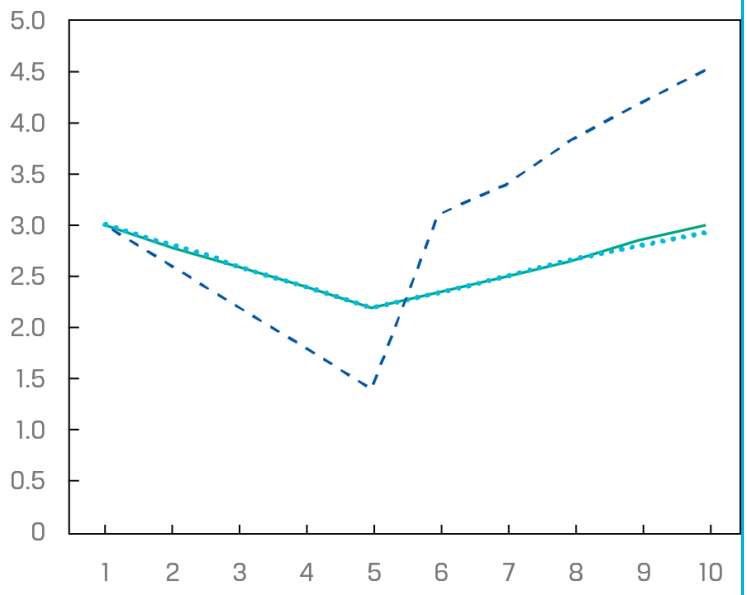
Question Set

1. The chart below shows a spot curve and its implied forward curve. Identify the forward curve, and justify your choice.



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Hide Solution

Solution:

The forward curve is represented by the line that moves from 3.0% to approximately 1.5% in Year 5 and then slopes dramatically higher from Year 5 to 6, ending at about 4.5% in Year 10.

When the spot curve is sloping downward (Years 1–5), forward rates will be below spot rates. When the spot curve is sloping upward (Years 6–10), forward rates will be above spot rates.

For Questions 2 and 3, refer to the spot rate data shown in Exhibit 9. It is important to note that some bonds will trade at negative yield. This was prevalent among developed market sovereign bonds in the wake of the financial crisis as central banks pursued accommodative monetary policy in response to below-target inflation. Negative rates do not change the fundamental mathematics used in bond pricing.

Exhibit 9:

Negative Spot Rates (Germany and Switzerland)

Maturity	Germany	Switzerland
1-year	–0.6965%	–0.7882%
2-year	–0.7034%	–0.7133%
3-year	–0.7021%	–0.6435%
4-year	–0.6418%	–0.5615%
5-year	–0.5578%	–0.4757%
6-year	–0.4796%	–0.4080%

9-year	–0.2524%	–0.1945%
10-year	–0.1779%	–0.1216%

2. Calculate the implied one-year forward rate in nine years in Switzerland.

Hide Solution

Solution:

The question is requesting the 9y1y forward rate. This is calculated from the 10-year and 9-year spot rates, as follows:

$$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B$$

$$(1 + -0.001945)^9 \times (1 + IFR_{9,1})^1 = (1 + -0.001216)^{10}$$

$$IFR_{9,1} = 0.537\%$$

Thus, the implied one-year forward rate in nine years for Swiss government bonds is 0.537%.

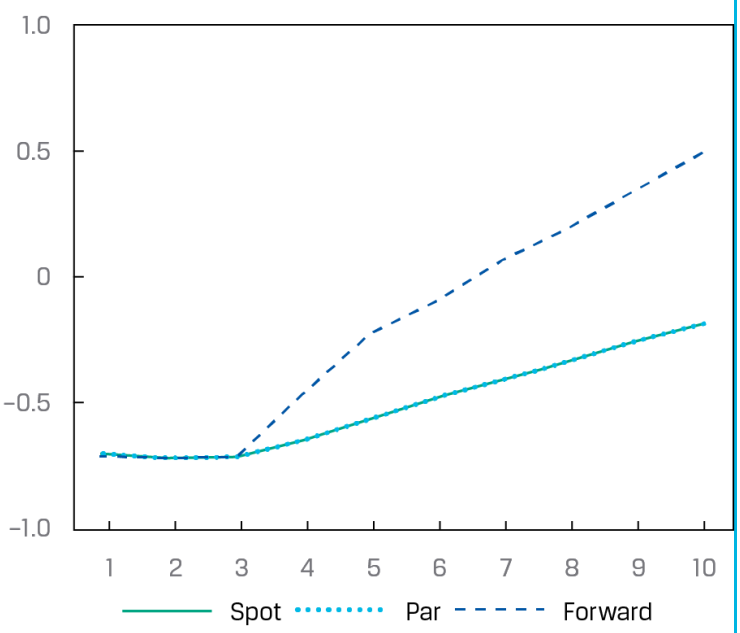
3. Explain why a country with negative spot rates, such as Switzerland, may have positive forward rates.

Hide Solution

Solution:

As shown in earlier examples, an upward-sloping spot curve (which Switzerland has) will be characterized by forward rates that are greater than spot rates. As the negative rates become less negative and move closer to zero, the more likely that forward rates will be positive with an upward-sloping spot curve.

4. The following chart shows spot and forward curves for German sovereign bonds. Explain why the forward curve is close to identical to the spot curve for two and three-year maturities.



Hide Solution

Solution:

In flat term structure environments, forward rates equal spot rates. The spot rates in Germany for two and three-year maturities appear to be approximately

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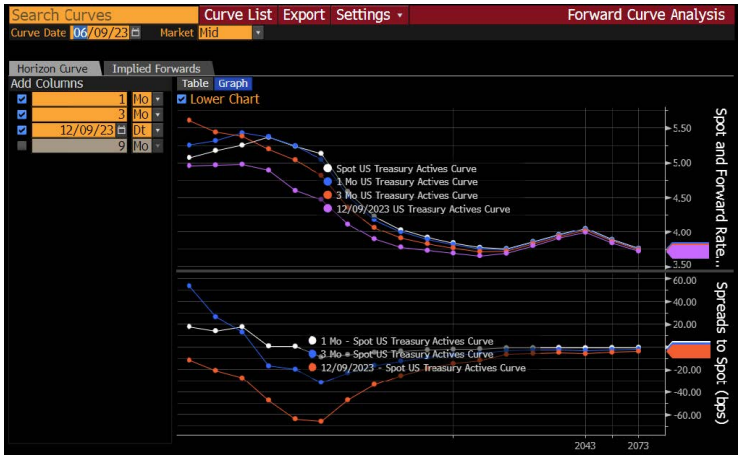
{FWCV} FORWARD CURVE ANALYSIS

The Bloomberg FWCV function enables you to access the forward rate model used to examine the direction that interest rates will move in the future.

> "FWCV <GO> allows you to analyze forward rates for global swap curves, corporate bond curves, and government bond curves at multiple horizon dates, so you can anticipate and compare the direction in which the interest rates will move over time." -Bloomberg

To explore this function, type FWCV into the Bloomberg command line and press Enter or <GO>. Type "US Treasury Actives Curve" (or "I25") into the Search Curves amber field at the top left, and select the US Treasury Actives Curve from the dropdown. Click from the Table view to the Graph view using the gray buttons above the table. The chart shows where the yield curve will be in future according to the model. By changing the horizon curve date at the top left, you can alter the date of the projected future yield curve.



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GF The forward curve is the dashed line because it was the dashed line in all of the samples... jk but funny in my head

Created 7 months ago by Gregory Fisher

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